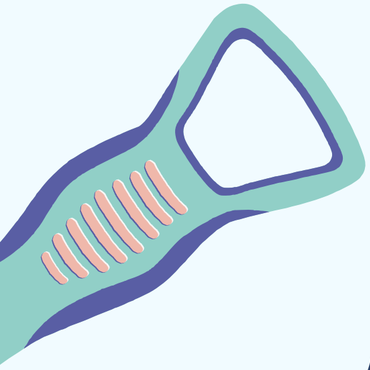




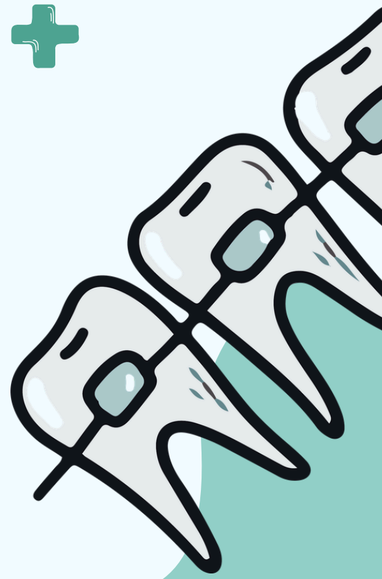


CLASS IV CAVITY AND RESTORATION

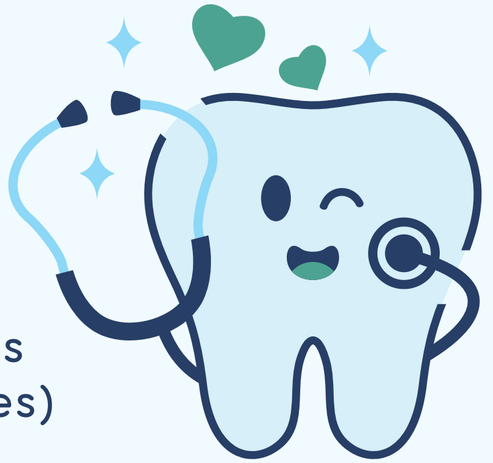


A Full Guide





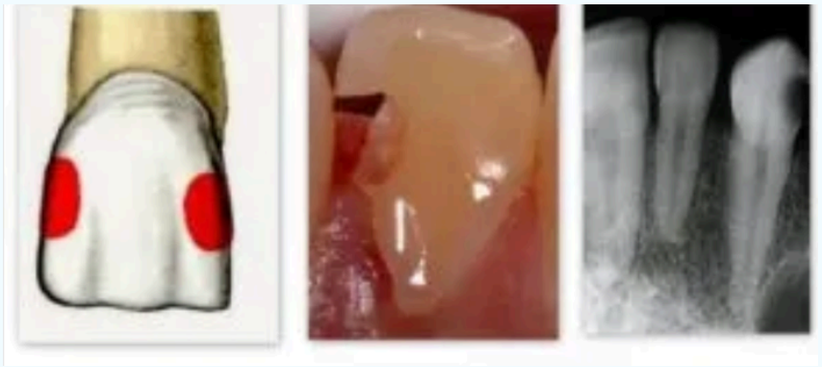
INTRODUCTION



What is a class IV cavity?

According to GV Black classification, class IV cavities refer to lesions involving the proximal surfaces of anterior teeth (incisors and canines) that **include** the incisal edge.

While Class III cavities include **the proximal surface ONLY** without including the incisal edge.



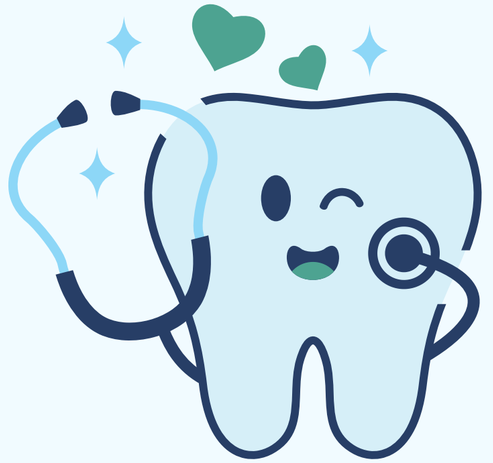
VS



CLASS III

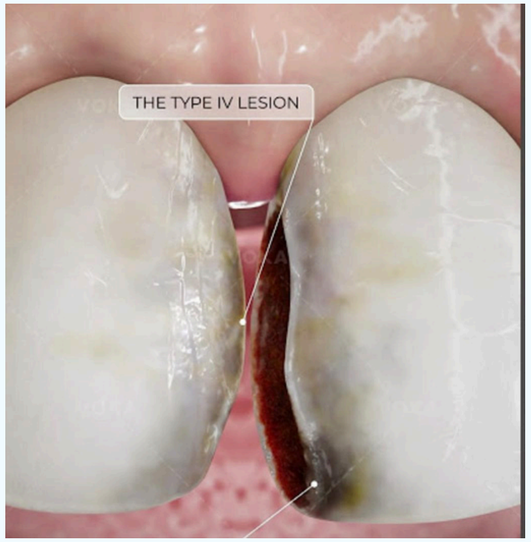
CLASS IV

INTRODUCTION



Class IV lesions are caused mainly by two factors:

- Caries involving **proximal sufaces + incisal edges.**
- Trauma (can involve the pulp sometimes and require endodontic treatment before restoration.

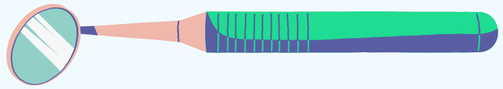


CARIES CLASS IV



TRAUMA CLASS IV

CLASS IV CAVITY PREP



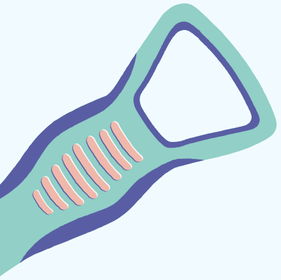
In case the class IV Cavity is caused by **Caries**:

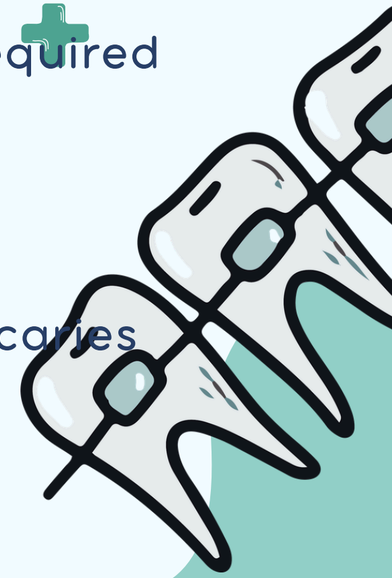
- All caries must be removed first
- If pulp exposure occurs, then endodontic treatment must be done before any further steps.
- Next, any sharp or undermined enamel must be removed.
- Then a bevel on the facial surface is done.



In case the class IV Cavity is caused by **Trauma**:

- The tooth and soft tissues should be cleaned properly from any blood.
- The mobility of the tooth should be assesed. If it needs splinting do it.
- Look for pulp exposure as well, if present then endodontic treatment is required first.
- Next, any sharp or undermined enamel must be removed.
- Then a bevel on the facial surface is done.





However, in preclinical setting, our acrylic tooth is completely sound with no caries pattern or trauma to follow.
So, What to do?
Check out the next slide to find out!





BEFORE STARTING YOUR PREP...

You need to prepare an impression of the tooth you're using and its neighbouring tooth. This should be done while the tooth is **still sound** before doing anything to it.

Why is this done?

The impression will be later used during composite restoration to help you build up the tooth with the correct form and **restore it to its original shape.**
The material used for this step is **condensation silicone.**
The exact process will be shown in a video in the next slide.



INDEX TECHNIQUE



CLASS IV CAVITY PREP

For our preclinical practice purposes, follow the following method:

- Using a pencil, draw a diagonal line from the incisal edge to the middle third of the tooth.
- Reduce the tooth structure below that line to mimic the form of a fractured tooth.
- Remove any sharp or undermined margins
- Then start working on the facial surface bevel.



PHOTO SHOWS THE FRACTURE LINE TO BREAK THE TOOTH TO MIMIC TRAUMA

SHOWN IN GREEN IS THE PART TO BE REMOVED

CLASS IV CAVITY PREP



What is the purpose of the Facial Surface Bevel?

- Creates space for the restoration so that the restored tooth does not appear or feel “bulky”
- Allows the composite restoration to blend in seamlessly with the rest of the tooth
- Prevent the presence of a demarcation/separation line between tooth and restoration.





BLUE COLOR SHOWING THE POSITION OF THE BEVEL: ABOVE FRACTURE LINE AND CONTAINED IN MIDDLE THIRD ONLY

CLASS IV CAVITY PREP



What You Need for the Perfect Restoration

Micromotor





Low speed contra



Composite enamel and dentin shades



Matrix band



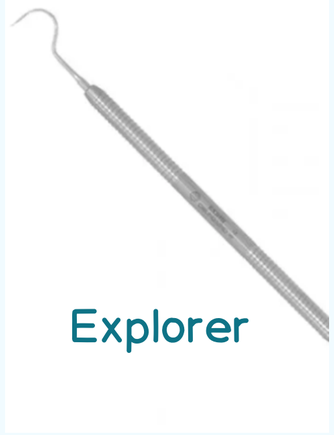
Lightcure



Plastic Instrument



Ball Burnisher



Explorer



Microbrush

Wedges





Polishing kit



Finishing Discs







Stones

Isolation

In a clinical setting, proper isolation is key to a succsessful restoration.

Composite restorations are **especially hydrophobic (ie sensitive to moisture)**. So your working field should be isolated properly from moisture in the oral cavity like saliva and blood.

Isolation can be done using several methods depending on each case.

Rubber Dam isolation is considered the **GOLDEN STANDARD** of isolation.

However, in some cases, rubber dam application is not possible .

Alternatives to rubber dam can be :

- Cotton roll isolation
- Isolite system
- Combination of bite blocks, high volume suction, cheeck retractors.









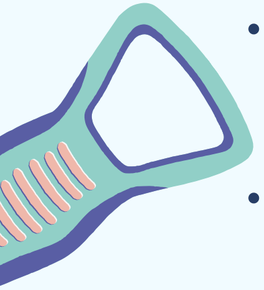
CLASS IV RESTORATION

Your technique, step by step: (think of it as an art project and let your inner artist shine!)

- Remember the impression you made before prepping the tooth? You will now use it to restore the palatal surface of the tooth , creating a **very thin layer of composite called "Palatal Shell"**



- The palatal shell is made using the enamel shade of composite.



- Next assemble your matrix bands and wedges properly to restore the **proximal surfaces also using enamel shade of composite.**
- Now you have created the borders of your tooth , you can now remove the matrix and work free-hand
- Start layering the facial surface of the tooth **starting at the bevel using dentin shade of composite.**



- Repeat this step several times till you reach proper thickness and contour.



- The last layer of the facial surface willl be done using enamel shade



- Detailed finishing and polishing should be done to restore the anatomy and ensure a smooth esthetically pleasing restoration.

PALATAL SHELL



PROXIMAL SURFACE



FACIAL LAYERING



FINISHING AND POLISHING



HOW TO CHECK YOUR WORK?



Check that the tooth is not bulky on any surface.	Check using probe that there is no excess composite remaining	check that your restoration is properley finished and polished and there is no roughness.
Check that surface anatomy like ridges and mamelons are restored	Check that there is no separation line between tooth and restoration	
Check that incisal edge thickness is “knife edge”	Using probe, check that your restoration has sealed all margins.	
Check that the contact is properly restored	Check that the restoration restored the natural contour of the tooth.	



HOW TO CHECK YOUR WORK?



CONGRATS! YOU HAVE NOW MASTERED CLASS IV CAVITY AND RESTORATION!



Contact Us!



If you have any further questions, stuck in any step or want to show us your progress, feel free to contact us completely free of charge!

Our talented instructor Dr. Salma is waiting for you!

Want to book a one on one session? Just give us a DM!



TikTok: @dentahelp

https://www.tiktok.com/@dentahelp?_r=1&_t=ZS-97oeLMuArIr

Whatsapp: +201105723303









THANK YOU

